

Practical 1

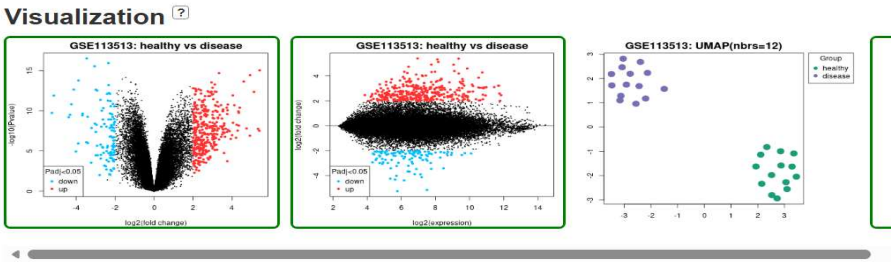
a: Find the gene expression dataset GSE113513 from NCBI and describes its features (Platform, BioProject No., case/control samples, No. of genes, disease name, tissue types, country names, PubMed-ID, publication date, name of organisms).

Answer:

Feature	Description
Dataset ID	GSE113513
Title	<i>Gene expression profile of human colorectal carcinoma</i>
Database	NCBI GEO (Gene Expression Omnibus)
Platform	GPL15207 — Affymetrix PrimeView Human Gene Expression Array
BioProject No.	PRJNA451417
Disease Name	Colorectal Cancer
Organism	Homo sapiens
Experiment Type	Expression profiling by array
Case Samples	14 colorectal carcinoma tissue samples
Control Samples	14 matched non-cancerous surrounding tissue samples
Total Samples	28
Tissue Type	Colorectal carcinoma tissue and adjacent non-cancerous colorectal tissue
Number of Genes	1,868 differentially expressed genes (778 upregulated, 1,090 downregulated) reported in the study
Country	China
Institution	Fujian University of Traditional Chinese Medicine
PubMed ID	35096810
Publication Date	Dataset released publicly on April 24, 2018
Submission Date	April 23, 2018
Last Update	February 2, 2022

b: Analyze the dataset GSE113513 by using LIMMA through GEO2R web tool and display the results and draw the volcano plot.

Answer:



Top differentially expressed genes [?](#)

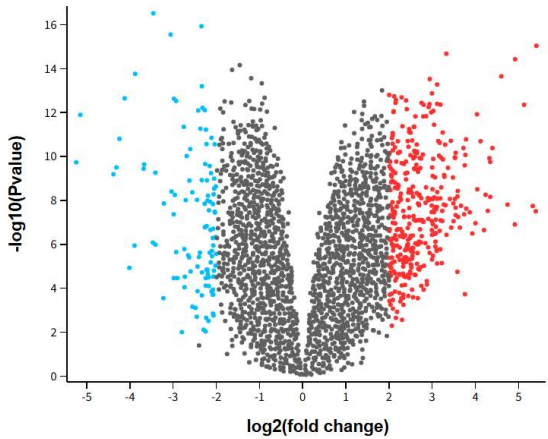
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ID	adj.P.Val	P.Value	t	B	logFC
▶ 11739128_a...	1.50e-12	3.04e-17	-18.18	28.9	-3.457
▶ 11719434_a...	2.87e-12	1.16e-16	-17.27	27.6	-2.342
▶ 11731795_at	4.66e-12	2.83e-16	-16.7	26.8	-3.056
▶ 11737294_a...	1.12e-11	9.09e-16	15.96	25.7	5.413
▶ 11734322_at	2.05e-11	2.08e-15	15.45	25	3.328
▶ 11732838_at	3.06e-11	3.71e-15	15.11	24.4	4.92
▶ 11718027_a...	4.86e-11	6.89e-15	-14.74	23.8	-1.454
▶ 11757706_a...	7.09e-11	1.15e-14	-14.45	23.3	-1.633
▶ 11728234_a...	9.47e-11	1.73e-14	-14.22	23	-3.877
▶ 11731957_at	1.10e-10	2.23e-14	14.07	22.7	4.601

Volcano Plot:

Volcano plot
GSE113513: Gene expression profile of human
colorectal carcinoma
healthy vs disease, Padj<0.05

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Select contrast to display

☒ healthy vs disease

[Download significant genes](#)

c: Identify the differentially expressed genes (DEGs) using the cutoff P-value>0.05 and $|\log_2FC| > 2.0$. Separate the upregulated and downregulated DEGs.

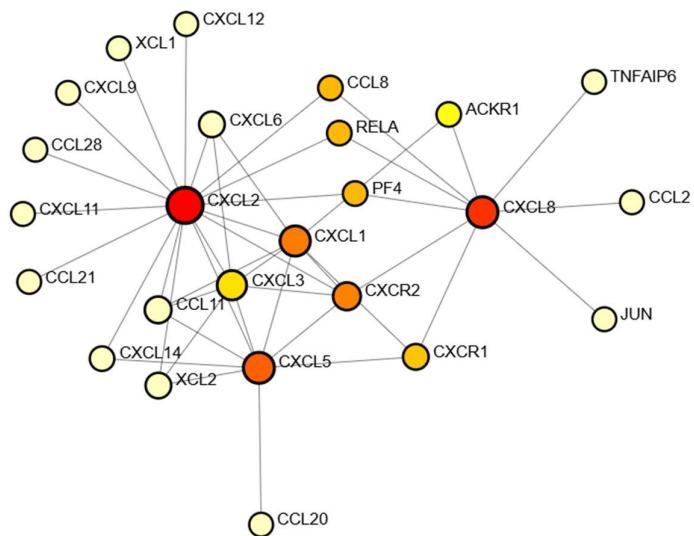
Answer:

Down-regulated genes:

ID	Gene.Symbol	Gene.Title	log2(fold change)	-log10(Pvalue)
11744030_a_at	MMP7	matrix metalloproteinase 7	-5.243	9.737
11758028_s_at	FOXQ1	forkhead box Q1	-5.147	11.897
11756074_a_at	MMP7	matrix metalloproteinase 7	-4.38	9.195
11744031_at	MMP7	matrix metalloproteinase 7	-4.311	9.502
11722783_at	FOXQ1	forkhead box Q1	-4.241	10.806
11728232_a_at	CLDN1	claudin 1	-4.12	12.647
11744424_a_at	KRT23	keratin 23, type I	-4.011	4.935
11756343_a_at	DPEP1	dipeptidase 1 (renal)	-3.891	5.941
11728234_a_at	CLDN1	claudin 1	-3.877	13.763
11717301_at	TACSTD2	tumor-associated calcium signal transducer 2	-3.681	9.442
11735833_a_at	CEMP1	cell migration inducing protein, hyaluronan-binding	-3.669	9.64
11743240_a_at	LY6G6D	lymphocyte antigen 6 complex, locus D	-3.468	6.084
11739128_a_at	CDH3	cadherin 3, type 1, P-cadherin (placental)	-3.457	16.518
11745361_a_at	CRNDE	colorectal neoplasia differentially expressed	-3.408	9.257
11718841_s_at	CXCL8	chemokine (C-X-C motif) ligand 8	-3.406	5.986
11719876_at	CKMT2	creatine kinase, mitochondrial 2 (sarcomeric)	-3.225	3.545
11728476_a_at	CXCL3	chemokine (C-X-C motif) ligand 3	-3.211	7.861
11731795_at	EPHX4	epoxide hydrolase 4	-3.056	15.549
11720372_at	TESC	tescalcin	-3.036	8.408
11728477_at	CXCL3	chemokine (C-X-C motif) ligand 3	-2.984	7.37
11754026_a_at	CXCL8	chemokine (C-X-C motif) ligand 8	-2.983	4.462

Up-regulated Genes:

ID	Gene.Symbol	Gene.Title	log2(fold change)	-log10(Pvalue)
11737294_a_at	TMIGD1	transmembrane and immunoglobulin domain	5.413	15.042
11729583_x_at	CA1	carbonic anhydrase I	5.399	7.507
11729582_s_at	CA1	carbonic anhydrase I	5.329	7.75
11723302_a_at	CHGA	chromogranin A	5.128	12.351
11732838_at	GUCA2B	guanylate cyclase activator 2B (uroguanylin)	4.92	14.43
11728771_a_at	CLCA4	chloride channel accessory 4	4.913	6.906
11742762_at	GCG	glucagon	4.745	7.805
11731957_at	GUCA2A	guanylate cyclase activator 2A (guanylin)	4.601	13.652
11717820_a_at	SLC4A4	solute carrier family 4 (sodium bicarbonate cotransporter)	4.394	10.384
11726764_at	AQP8	aquaporin 8	4.345	8.163
11746898_x_at	ADH1B	alcohol dehydrogenase 1B (class I), beta	4.343	9.752
11720819_s_at	CXCL12	chemokine (C-X-C motif) ligand 12	4.325	9.922
11722059_at	ZG16	zymogen granule protein 16	4.288	7.527
11729831_a_at	CLDN8	claudin 8	4.237	8.259
11733744_at	MS4A12	membrane-spanning 4-domains, subfamily A	4.2	6.648
11721557_a_at	ABCA8	ATP binding cassette subfamily A member 8	4.119	10.695
11729832_s_at	CLDN8	claudin 8	4.045	8.509
11732094_a_at	PYY	peptide YY	4.037	11.923
11730765_at	CD177	CD177 molecule	4.002	6.965
11750303_x_at	B3GALT5-AS1	B3GALT5 antisense RNA 1	3.929	6.497
11733115_a_at	CA2	carbonic anhydrase II	3.87	7.46

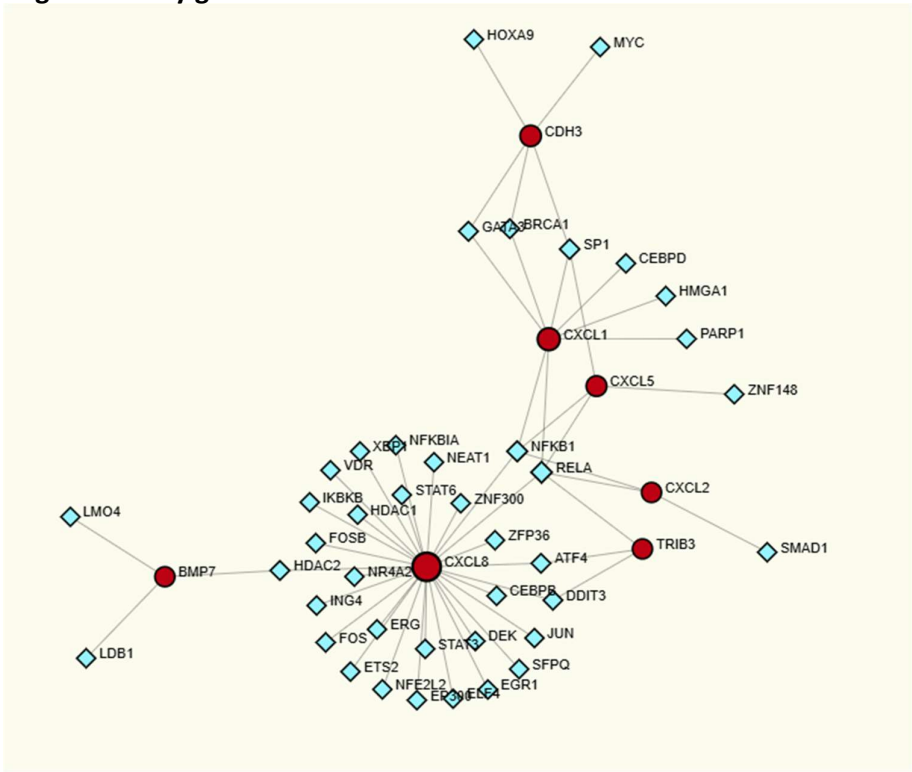


Top 10 Key genes by using degree and betweenness:

	A	B	C	D	E	F	G	H
1	Id	Label	Degree	Betweenness	Expression	Rank(D)	Rank(E)	Average Rank
2	2920	CXCL2	17	165.97	0	1	1	1
3	655	BMP7	14	133.5	0	2	2	2
4	3576	CXCL8	9	76.33	0	3.5	3	3.25
5	6374	CXCL5	9	37.69	0	3.5	8	5.75
6	2550	GABBR1	8	28	0	6	9.5	7.75
7	57761	TRIB3	8	28	0	6	9.5	7.75
8	1001	CDH3	6	67	0	11.5	4	7.75
9	9076	CLDN1	6	55	0	11.5	5	8.25
10	2919	CXCL1	8	25.12	0	6	11	8.5
11	3624	INHBA	6	49.5	0	11.5	6	8.75
12	9319	TRIP13	7	21	0	8.5	14	11.25
13	3579	CXCR2	5	22.48	0	14.5	13	13.75
14	8549	LGR5	6	15	0	11.5	18	14.75
15	25902	MTHFD1L	5	10	0	14.5	20	17.25
16	2921	CXCL3	7	4.89	0	8.5	30	19.25
17	4316	MMP7	4	9.33	0	17	21.5	19.25
18	4314	MMP3	4	9.33	0	17	21.5	19.25
19	7082	TIP1	2	48	0	37	7	22

(e) Construct the TFs-KGs interaction network and identify 5 key TFs as the transcription regulators of KGs.

Answer: Using the 10 Key genes here we have constructed the TFs-KGs interaction network.



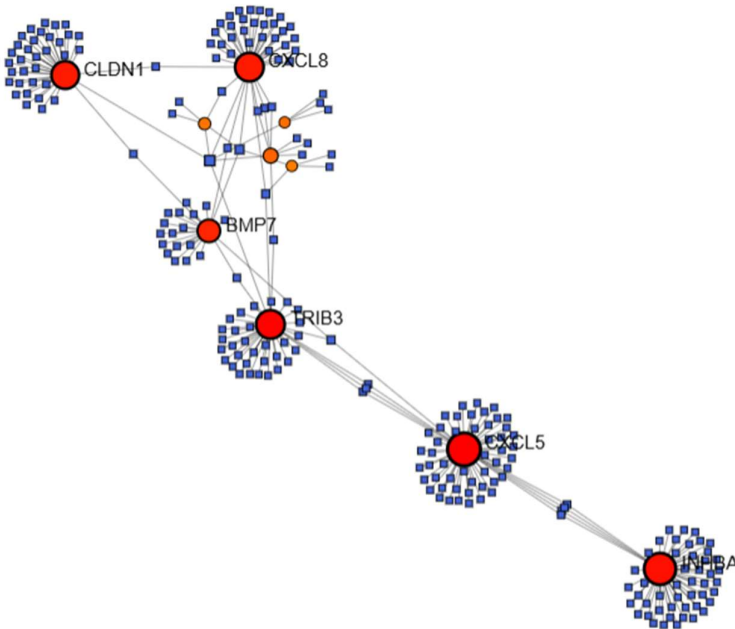
Top 5 key TFs as the transcription regulators of KGs: (Highlighted in red color)

	A	B	C	D	E	F	G	H
1	Id	Label	Degree	Betweenness	Expression	Rank(D)	Rank(B)	Average
2	3576	CXCL8	29	926.92	0	1	1	1
3	2919	CXCL1	8	310.92	0	2	2	2
4	5970	RELA	5	261.67	0	3.5	3	3.25
5	1001	CDH3	5	93.5	0	3.5	6	4.75
6	4790	NFKB1	4	228	0	5.5	4	4.75
7	6374	CXCL5	4	91.42	0	5.5	7	6.25
8	655	BMP7	3	91	0	8.5	8	8.25
9	6667	SP1	3	68.67	0	8.5	9	8.75
10	3066	HDAC2	2	132	0	13	5	9
11	2920	CXCL2	3	46.42	0	8.5	10	9.25
12	57761	TRIB3	3	10.83	0	8.5	13	10.75
13	672	BRCA1	2	31	0	13	11.5	12.25
14	2625	GATA3	2	31	0	13	11.5	12.25
15	468	ATF4	2	9.83	0	13	14.5	13.75
16	1649	DDIT3	2	9.83	0	13	14.5	13.75

(f) Construct the miRNA-KGs interaction network and identify 5 key miRNA as the post-transcription regulators of KGs.

Answer:

miRNA-KGs interaction network:



Here are 5 key miRNA as the post-transcription regulators of KGs: (marked red)

	A	B	C	D	E	F	G	H
1	Id	Label	Deg	Between	Exp	Ran	Ra	Ave
2	6374	CXCL5	57	18394.78	0	1	1	1
3	3624	INHBA	51	10437	0	2	3	2.5
4	57761	TRIB3	39	15122.42	0	3.5	2	2.75
5	3576	CXCL8	39	7838.59	0	3.5	5	4.25
6	9076	CLDN1	36	7675.37	0	5	6	5.5
7	MIMAT0000765	hsa-miR-335-5p	6	8952.58	0	8	4	6
8	655	BMP7	22	6075.35	0	6	8	7
9	MIMAT0016915	hsa-miR-4284	3	6749.73	0	13.5	7	10.25
10	MIMAT0000422	hsa-miR-124-3p	3	2490.36	0	13.5	12	12.75
11	2550	GABBR1	8	1015.47	0	7	19	13
12	MIMAT0000416	hsa-miR-1-3p	4	1181.29	0	11	18	14.5
13	1001	CDH3	4	732	0	11	20	15.5
14	2920	CXCL2	5	615.35	0	9	23	16
15	MIMAT0005592	hsa-miR-1237-3p	2	2674.1	0	22.5	10	16.25
16	MIMAT0018988	hsa-miR-4464	2	2674.1	0	22.5	10	16.25
17	MIMAT0019884	hsa-miR-4748	2	2674.1	0	22.5	10	16.25
18	2919	CXCL1	4	499.68	0	11	24	17.5

(g) Construct the diseases-KGs interaction network and identify top 10 diseases risks as comorbidities associated with KGs.

Answer: Top 10 diseases from Enrichr (DisGeNET database)

DisGeNET
Bar Graph
Table
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Hover each row to see the overlapping genes.

10 entries per page
Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	Peritonsillar Abscess	5.033e-9	0.000005104	1713.00	32730.63
2	Endometriosis	1.093e-8	0.000005540	60.53	1109.60
3	Chronic Airflow Obstruction	4.872e-7	0.0001647	285.14	4144.43
4	Dental caries	0.000001017	0.0002577	79.61	1098.59
5	Mammary Neoplasms	0.000001466	0.0002733	29.61	397.75
6	Ulcerative Colitis	0.000001617	0.0002733	31.49	419.87
7	Degenerative polyarthritis	0.000002359	0.0003417	29.41	381.11
8	Neoplasm Metastasis	0.000003491	0.0003871	37.00	464.93
9	Neutrophilia (disorder)	0.000003518	0.0003871	142.36	1787.65
10	Periodontitis	0.000003817	0.0003871	56.53	705.26

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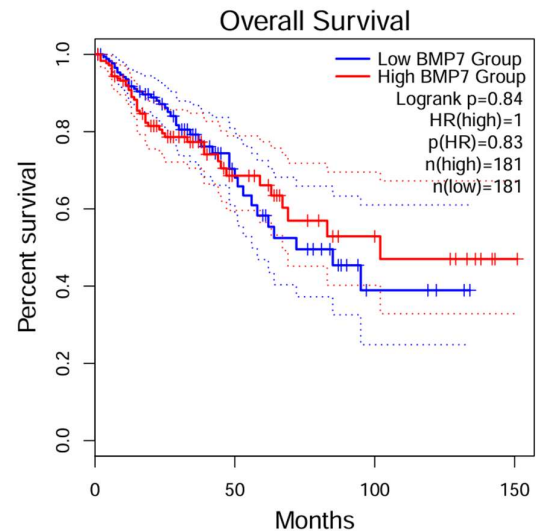
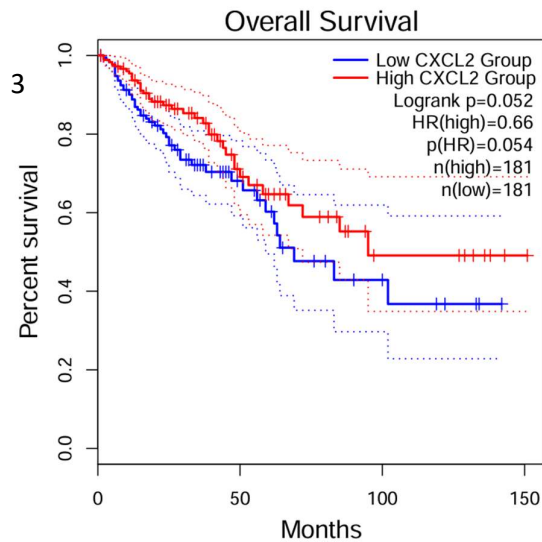
Top 10 diseases using NetworkAnalyst (DisGeNET database) (top 10 diseases marked in RED)

1	Id	Label	Degr	Betw	Expre	Ra	Rar	Averag
2	3576	CXCL8	55	7738.3	0	1	1	1
3	1001	CDH3	54	5989	0	2	2	2
4	2920	CXCL2	19	1613.3	0	3	5	4
5	2919	CXCL1	11	1618.8	0	4	4	4
6	C0024117	Chronic Obstructive Airway Disease	3	989.14	0	7	6.5	6.75
7	C0035126	Reperfusion Injury	3	989.14	0	7	6.5	6.75
8	2550	GABBR1	5	550	0	5	9	7
9	C0009324	Ulcerative Colitis	2	4644	0	14.5	3	8.75
10	57761	TRIB3	3	277	0	7	11	9
11	C0001973	Alcoholic Intoxication, Chronic	2	675	0	14.5	8	11.25
12	C0023893	Liver Cirrhosis, Experimental	2	411	0	14.5	10	12.25
13	C0003873	Rheumatoid Arthritis	2	197.14	0	14.5	14	14.25
14	C0008370	Cholestasis	2	197.14	0	14.5	14	14.25
15	C0021368	Inflammation	2	197.14	0	14.5	14	14.25
16	C0034069	Pulmonary Fibrosis	2	197.14	0	14.5	14	14.25
17	C1458155	Mammary Neoplasms	2	197.14	0	14.5	14	14.25
18	C0020517	Hypersensitivity	2	69	0	14.5	17.5	16
19	C0032285	Pneumonia	2	69	0	14.5	17.5	16
20	C0036982	Shock, Hemorrhagic	2	65	0	14.5	19	16.75
21	6374	CXCL5	2	65	0	14.5	20	17.25

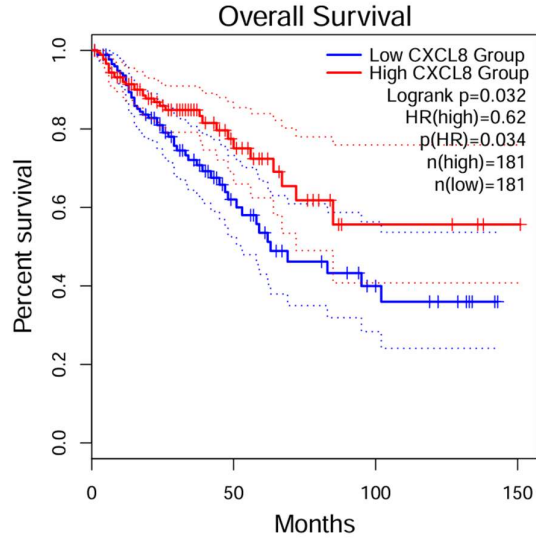
(h) Perform survival analysis with KGs and interpret the results.

Answer: 1. CXCL2

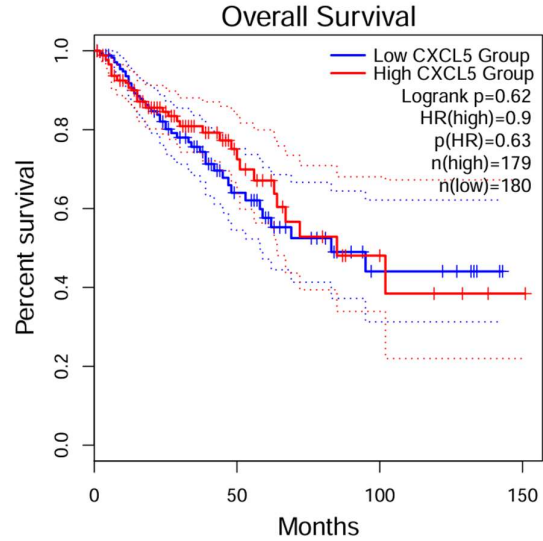
2. BMP7



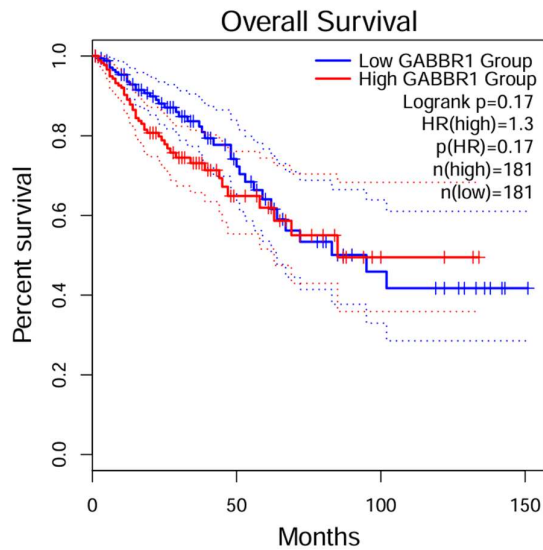
3. CXCL8



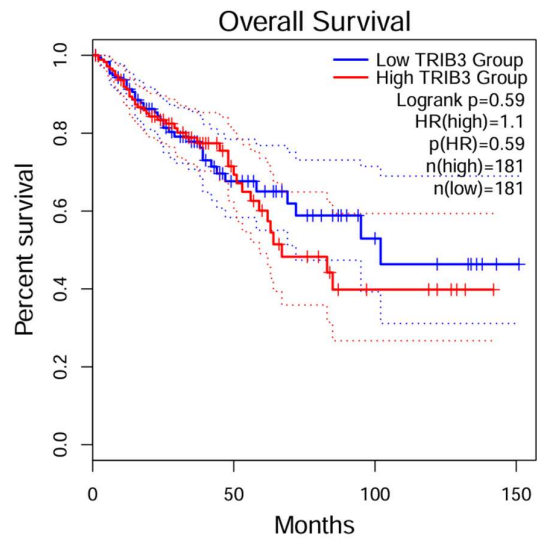
4. CXCL5



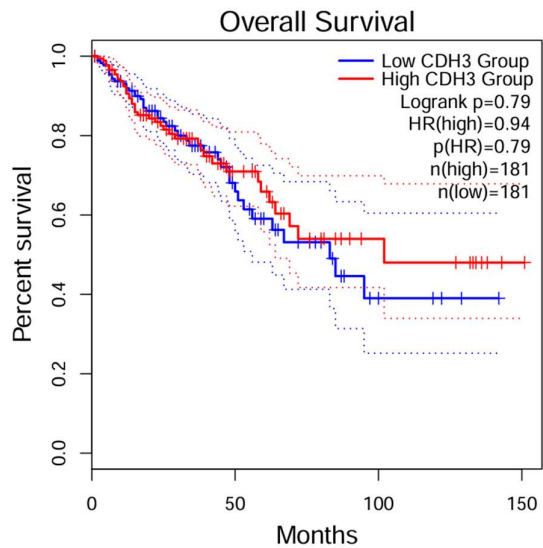
5. GABBR1



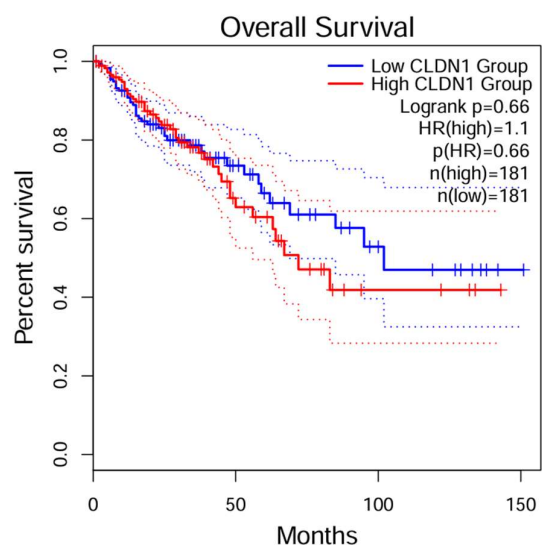
6. TRIB3



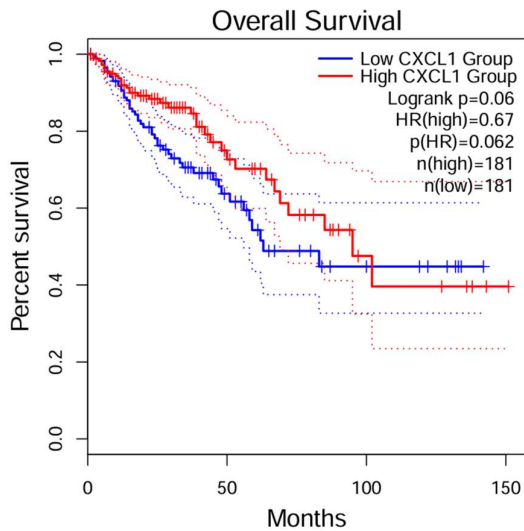
7. CDH3



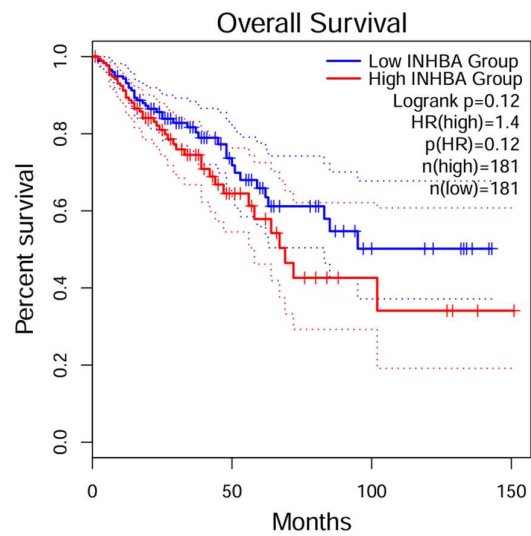
8. CLDN1



9. CXCL1



10. INHBA



(i) Identify KGs related drugs for the treatment against the disease.

Answer: Using Enrichr:

DSigDB

Bar Graph

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Hover each row to see the overlapping genes.

10 entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	depropine PC3 UP	2.559e-13	2.245e-10	482.13	13978.78
2	STOCK1N-35215 PC3 UP	5.536e-13	2.427e-10	420.82	11876.67
3	pizotifen PC3 UP	1.604e-11	4.689e-9	463.88	11530.26
4	clomipramine PC3 UP	1.065e-10	2.335e-8	168.87	3877.72
5	CP-690334-01 PC3 DOWN	1.520e-10	2.437e-8	832.25	18814.57
6	thioridazine PC3 UP	1.667e-10	2.437e-8	156.32	3519.39
7	1-NITROPYRENE CTD 00001569	4.858e-10	6.086e-8	130.01	2788.16
8	chloropyramine PC3 UP	1.451e-9	1.413e-7	443.56	9026.96
9	promethazine PC3 UP	1.451e-9	1.413e-7	443.56	9026.96
10	8-azaguanine PC3 UP	3.369e-9	2.954e-7	93.09	1816.07

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DGIdb Drug Targets 2024

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10 entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	CYCLOPHOSPHAMIDE	0.0006453	0.02259	66.38	487.64
2	BACLOFEN	0.004492	0.03076	277.53	1500.17
3	DACARBAZINE	0.004990	0.03076	246.68	1307.49
4	LEFLUNOMIDE	0.004990	0.03076	246.68	1307.49
5	NAPROXEN	0.005985	0.03076	201.81	1032.95
6	HYDROQUINONE	0.007476	0.03076	158.54	776.21
7	DANAZOL	0.007476	0.03076	158.54	776.21
8	METHIMAZOLE	0.007973	0.03076	147.96	714.91
9	RADIOTHERAPY	0.01144	0.03076	100.85	450.83
10	FENRETINIDE	0.01144	0.03076	100.85	450.83

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(j) Perform GO and KEGG pathway enrichment analysis of DEGs and identify top 5 GO-term from each of Biological Process (BP), Molecular Function (MF) & Cellular Component (CC) and top 5 KEGG pathways.

Answer:

- Top 5 GO terms from Biological Process (BP):

GO Biological Process 2026

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Hover each row to see the overlapping genes.

10 entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	Antimicrobial Humoral Immune Response Mediated by Antimicrobial Peptide (GO:0061844)	3.062e-7	0.00008360	108.57	1628.41
2	Antimicrobial Humoral Response (GO:0019730)	7.588e-7	0.0001036	85.87	1210.05
3	Inflammatory Response (GO:0006954)	0.000009451	0.0008601	44.66	516.71
4	Negative Regulation of Cellular Process (GO:0048523)	0.0001048	0.007156	23.70	217.13
5	Positive Regulation of SMAD Protein Signal Transduction (GO:0060391)	0.0001737	0.009486	131.26	1136.48

- Top 5 **GO** terms from **Molecular Function (MF)**:

GO Molecular Function 2026

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Hover each row to see the overlapping genes.

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Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	Cytokine Activity (GO:0005125)	2.187e-10	5.686e-9	149.18	3318.23
2	Chemokine Activity (GO:0008009)	7.176e-9	8.532e-8	289.04	5420.31
3	Chemokine Receptor Binding (GO:0042379)	9.844e-9	8.532e-8	265.87	4901.62
4	Growth Factor Activity (GO:0008083)	0.00001738	0.0001130	81.95	898.15
5	CXCR Chemokine Receptor Binding (GO:0045236)	0.00002690	0.0001399	356.71	3753.85

- Top 5 **GO** terms from **Cellular Component (CC)**:

GO Cellular Component 2026

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Hover each row to see the overlapping genes.

entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	G Protein-Coupled Receptor Dimeric Complex (GO:0038037)	0.003495	0.06641	370.07	2093.27
2	Cell-cell Junction (GO:0005911)	0.01158	0.09751	14.71	65.59
3	Catenin Complex (GO:0016342)	0.01540	0.09751	73.93	308.54
4	Tertiary Granule Lumen (GO:1904724)	0.02717	0.1162	41.02	147.91
5	Specific Granule Lumen (GO:0035580)	0.03058	0.1162	36.30	126.60

Top 5 KEGG PATHWAYS:

KEGG 2026

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Hover each row to see the overlapping genes.

entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	CYTOKINE-CYTOKINE RECEPTOR INTERACTION	1.996e-9	1.058e-7	101.90	2041.20
2	RHEUMATOID ARTHRITIS	8.619e-8	0.000001536	150.77	2452.58
3	IL-17 SIGNALING PATHWAY	9.004e-8	0.000001536	149.07	2418.38
4	VIRAL PROTEIN INTERACTION WITH CYTOKINE AND CYTOKINE RECEPTOR	1.159e-7	0.000001536	139.61	2229.69
5	CHEMOKINE SIGNALING PATHWAY	0.000001618	0.00001715	70.60	941.38

PRACTICAL 2

(a) Download the gene expression dataset GSE22598 from NCBI and describes its features (Platform, BioProject No., case/control samples, No. of genes, disease name, tissue types, country names, PubMed-ID, publication date, name of organisms).

Feature	Description
Dataset Accession No.	GSE22598
Disease Name	Colorectal cancer (CRC)
Organism	Homo sapiens
Experiment Type	Expression profiling by array
Platform	GPL570 – Affymetrix Human Genome U133 Plus 2.0 Array
Approximate No. of Genes/Probes	~54,675 probe sets
BioProject No.	PRJNA128377
Case Samples	17 colorectal cancer tissues
Control Samples	17 adjacent non-cancerous colorectal tissues
Additional Samples	4 cell line samples (HT29 and HCT15; control and treated)
Total Samples	38
Tissue Type	Colorectal cancer tissue and adjacent normal colorectal tissue
Country	Japan
PubMed ID	21922135
Publication Date	2011
Submission Date to GEO	June 28, 2010
Main Study Title	“Clinical significance of UNC5B Expression in Colorectal Cancer”

(b) Analyze the dataset GSE22598 by using LIMMA r-package and display the results.

(c) Identify the differentially expressed genes (DEGs) using the cutoff $P\text{-value} > 0.05$ and $|\log_2FC| > 2.0$? Separate the upregulated and downregulated DEGs.

R code:

```
rm(list=ls(all=TRUE))  
library(limma)  
library(gclus)
```

```

library(Biobase)
library(dplyr)

getwd()
setwd("E:")
DataSet <- read.csv("GSE22598_series_matrix.csv", header = TRUE)

# Separate IDs
ID <- DataSet[,1]
DataS <- DataSet[,-1]

# Set rownames
rownames(DataS) <- ID

# Convert to matrix (NO voom needed)
Data <- as.matrix(DataS)

# Define groups (IMPORTANT: adjust if needed based on metadata!)
group <- factor(c(rep('Control', 17), rep('Case', 21)))

design.mat <- model.matrix(~ group)

fit <- lmFit(Data, design.mat)
fit <- eBayes(fit)

results <- topTable(fit, coef = 2, adjust = "BH",
                    n = nrow(Data), sort.by = "none")

head(results)

write.csv(results, "Limma_results_GSE22598.csv")

# Upregulated genes
Upregulated_data <- results %>%
  filter(adj.P.Val < 0.05 & logFC > 2)

write.csv(Upregulated_data,
          "Upregulated_GSE22598.csv")

# Downregulated genes
Downregulated_data <- results %>%
  filter(adj.P.Val < 0.05 & logFC < -2)

write.csv(Downregulated_data,
          "Downregulated_GSE22598.csv")

```

Upregulated DE Genes:

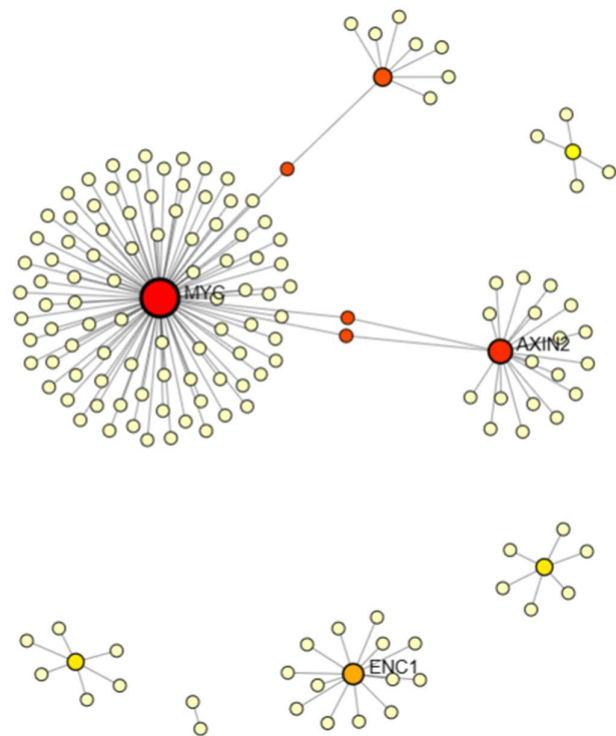
	A	B	C	D	E	F	G
1	ID	logFC	AveExpr	t	P.Value	adj.P.Val	B
2	1552296_at	3.681458	6.205024	9.677584	7.65E-12	2.73E-09	16.94334
3	1552365_at	3.225279	7.855174	7.20613	1.23E-08	6.65E-07	9.727516
4	1552367_a_at	2.143727	5.640971	8.352885	3.67E-10	4.32E-08	13.16448
5	1552870_s_at	2.079458	6.352753	3.512916	0.001153	0.006286	-1.37852
6	1553808_a_at	2.5677	5.906447	6.29716	2.15E-07	6.21E-06	6.929347
7	1553828_at	2.409075	7.186324	5.245605	6.01E-06	8.90E-05	3.678086
8	1555962_at	2.992629	7.478201	6.461336	1.28E-07	4.13E-06	7.437542
9	1555963_x_at	3.186109	7.713823	6.336571	1.89E-07	5.65E-06	7.051405
10	1557119_a_at	2.005944	7.838811	9.630843	8.73E-12	2.98E-09	16.81394
11	1561421_a_at	2.098195	8.533985	5.512931	2.58E-06	4.56E-05	4.500813
12	1563692_at	2.585647	7.00602	6.197408	2.94E-07	8.04E-06	6.620278
13	1568634_a_at	2.122211	7.103478	5.211981	6.68E-06	9.69E-05	3.57499
14	200795_at	2.595635	8.728584	4.179435	0.000163	0.001294	0.484675
15	201041_s_at	2.036791	11.47493	5.914119	7.23E-07	1.64E-05	5.742139
16	201058_s_at	2.12832	7.613043	3.409689	0.001543	0.007939	-1.65332
17	201137_s_at	2.0029	9.40548	3.767688	0.000554	0.003476	-0.68319
18	201325_s_at	2.38959	10.08854	6.267533	2.36E-07	6.72E-06	6.837569
19	201427_s_at	2.009195	11.77508	3.468997	0.001306	0.006944	-1.40595

Down-Regulated DE Genes:

	A	B	C	D	E	F	G
1	ID	logFC	AveExpr	t	P.Value	adj.P.Val	B
2	1552619_a_at	-2.16871	6.506845	-5.75219	1.21E-06	2.48E-05	5.240506
3	1553764_a_at	-2.93513	6.770681	-7.31976	8.64E-09	5.05E-07	10.07367
4	1554020_at	-2.13297	6.312104	-8.9313	6.60E-11	1.20E-08	14.84232
5	1554452_a_at	-2.24211	7.482996	-13.4034	4.74E-16	4.31E-12	26.28872
6	1555758_a_at	-2.08003	7.368427	-6.012	5.30E-07	1.28E-05	6.045559
7	1557910_at	-2.2463	9.239624	-7.36422	7.52E-09	4.56E-07	10.2088
8	1558028_x_at	-2.35137	7.359447	-10.4665	8.48E-13	5.15E-10	19.08184
9	1568609_s_at	-2.14101	6.062781	-4.98373	1.37E-05	0.000174	2.878084
10	200806_s_at	-2.09308	10.51422	-6.5476	9.72E-08	3.28E-06	7.704243
11	200832_s_at	-2.55071	9.768463	-7.98064	1.13E-09	1.04E-07	12.06366
12	201195_s_at	-2.78176	8.229493	-7.90638	1.42E-09	1.24E-07	11.84221
13	201291_s_at	-2.35502	9.108878	-6.1075	3.91E-07	1.00E-05	6.341587
14	201292_at	-2.23941	9.302331	-6.19575	2.96E-07	8.07E-06	6.61514
15	201340_s_at	-3.4123	8.685871	-19.4402	1.86E-21	1.01E-16	37.85634
16	201417_at	-2.2467	10.32403	-11.2463	1.04E-13	1.19E-10	21.11054
17	201506_at	-2.15588	10.26648	-4.10875	0.000202	0.001536	0.280786
18	201946_s_at	-2.2569	8.490037	-8.05049	9.15E-10	8.84E-08	12.2714

(d) Construct protein-protein interaction (PPI) network and identify top 10 key genes (KGs) from the network by using two topological measures degree and betweenness.

Answer: PPI Network:



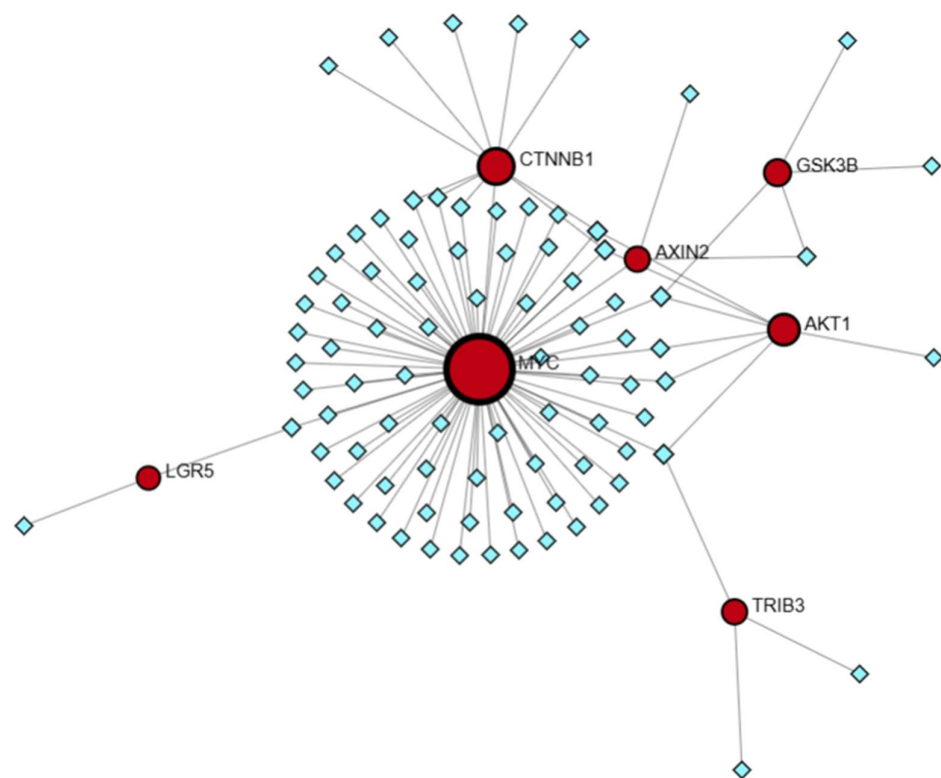
Top 10 Key Genes Using Both Degree and Betweenness:

Id	Label	Degree	Betweenness	Expression	Rank D	Rank B	Average
4609	MYC	91	6657.5	0	1	1	1
8313	AXIN2	20	1953.5	0	2	2	2
8507	ENC1	13	78	0	3	7	5
57761	TRIB3	8	798	0	4	6	5
2932	GSK3B	2	931	0	9	3.5	6.25
1499	CTNNB1	2	931	0	9	3.5	6.25
8549	LGR5	6	15	0	5.5	8.5	7
9076	CLDN1	6	15	0	5.5	8.5	7
207	AKT1	2	880	0	9	5	7
84962	AJUBA	4	6	0	7	10	8.5

(e) Construct the TFs-KGs interaction network and identify 5 key TFs as the transcription regulators of KGs.

Answer:

TFs-KGs Interaction Network:



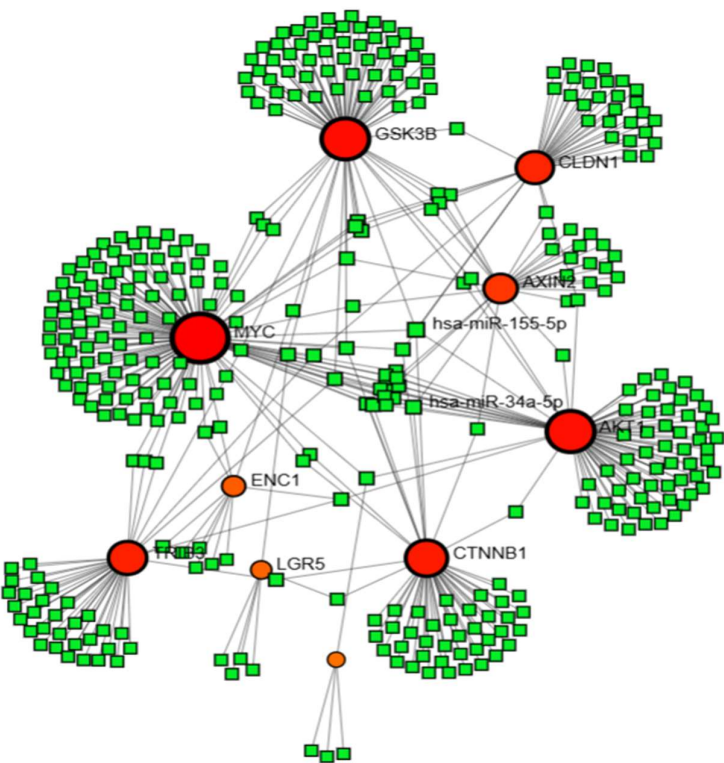
Top 5 Transcription Factors (Yellow Marked):

	A	B	C	D	E	F	G	H
1	Id	Label	Degree	Betweenness	Expression	Rank D	Rank B	Average
2	4609	MYC	75	4040.83	0	1	1	1
3	1499	CTNNB1	11	451.33	0	2	2	2
4	2932	GSK3B	4	184	0	4	5	4.5
5	5970	RELA	3	290.75	0	7.5	3	5.25
6	4790	NFKB1	3	283.75	0	7.5	4	5.75
7	207	AKT1	7	117.83	0	3	9	6
8	57761	TRIB3	3	181	0	7.5	6	6.75
9	8313	AXIN2	3	175	0	7.5	8	7.75
10	367	AR	3	30.5	0	7.5	11.5	9.5
11	7157	TP53	3	30.5	0	7.5	11.5	9.5
12	6925	TCF4	2	180	0	14.5	7	10.75
13	8549	LGR5	2	91	0	14.5	10	12.25
14	324	APC	2	23.75	0	14.5	13.5	14
15	6774	STAT3	2	23.75	0	14.5	13.5	14

(f) Construct the miRNA-KGs interaction network and identify 5 key miRNA as the posttranscription regulators of KGs.

Answer:

miRNA-KGs Interaction Network:



Top 5 miRNA:

	A	B	C	D	E	F	G	H
1	Id	Label	De	Between	Exp	Ran	Ran	Avg
2	4609	MYC	124	36980.3	0	1	1	1
3	207	AKT1	75	21660.7	0	2	3	2.5
4	2932	GSK3B	74	21883.2	0	3	2	2.5
5	1499	CTNNB1	50	15031.4	0	4	4	4
6	57761	TRIB3	39	11779.6	0	5	5	5
7	9076	CLDN1	36	10502.6	0	6	6	6
8	8313	AXIN2	27	6540.22	0	7	8	7.5
9	MIMAT0000646	hsa-miR-155-5p	5	7539.69	0	10	7	8.5
10	MIMAT0000255	hsa-miR-34a-5p	4	3392.4	0	11.5	9	10.3
11	8507	ENC1	10	2101.59	0	8	13	10.5
12	MIMAT0000083	hsa-miR-26b-5p	3	3197.34	0	19.5	10	14.8
13	MIMAT0000765	hsa-miR-335-5p	3	2620.9	0	19.5	11	15.3
14	8549	LGR5	7	1607.48	0	9	22	15.5
15	MIMAT0004945	hsa-miR-744-5p	3	2561.95	0	19.5	12	15.8
16	MIMAT0001080	hsa-miR-196b-5p	3	2023.54	0	19.5	14	16.8
17	MIMAT0000092	hsa-miR-92a-3p	3	2014.84	0	19.5	15	17.3

(g) Construct the diseases-KGs interaction network and identify top 10 diseases risks as comorbidities associated with KGs

Answer:

Top 10 Diseases:

DisGeNET

Bar GraphTableClustergramAppyter⚙️📘

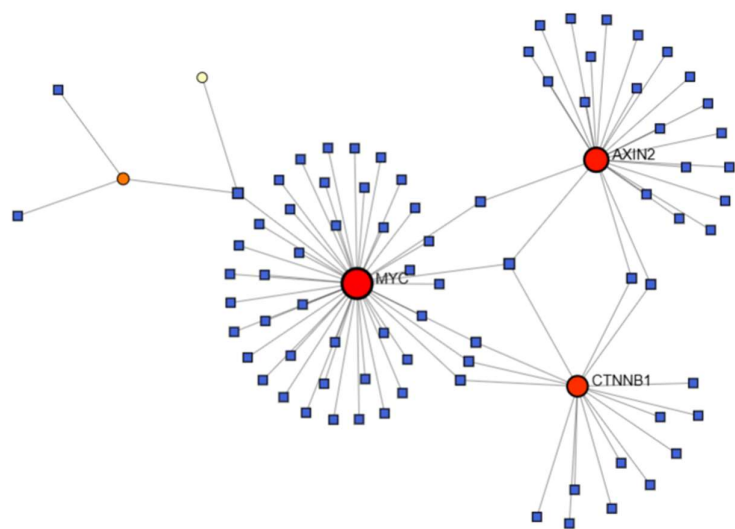
Hover each row to see the overlapping genes.

10 entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	Adenoma	1.564e-9	0.000002001	76.60	1553.26
2	Colorectal Carcinoma	4.509e-9	0.000002886	170690.00	3280180.73
3	Adenomatous Polyposis Coli	1.072e-8	0.000004549	76.18	1397.99
4	Colonic Neoplasms	1.422e-8	0.000004549	58.16	1050.96
5	Stomach Neoplasms	2.318e-8	0.000005935	54.00	949.30
6	Hepatoblastoma	5.155e-8	0.00001100	85.91	1441.68
7	Anaplasia	1.302e-7	0.00002089	70.91	1124.16
8	Colorectal Neoplasms	1.306e-7	0.00002089	41.42	656.60
9	Increased gastric cancer	1.586e-7	0.00002255	427.93	6700.13
10	Esophageal Neoplasms	1.920e-7	0.00002458	46.02	711.73

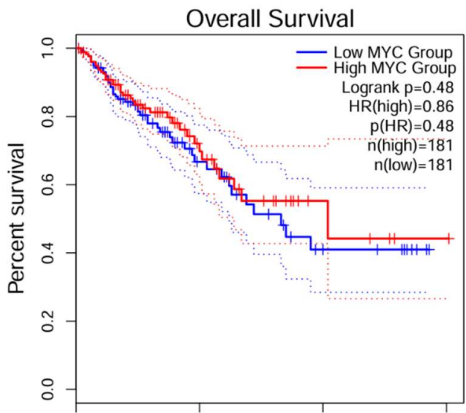
Diseases KGs Network:



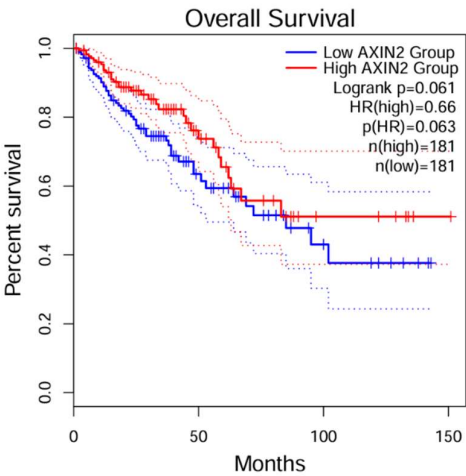
(h) Perform survival analysis with KGs and interpret the results.

Answer:

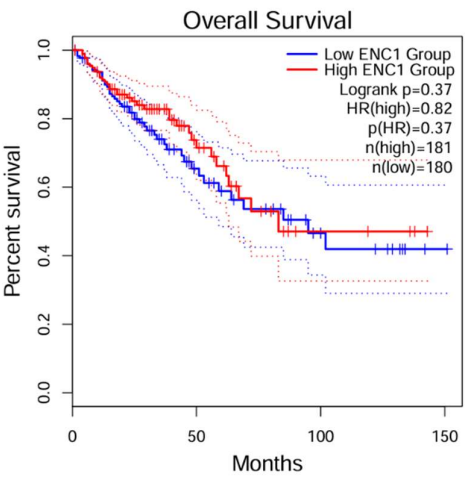
1. MYC



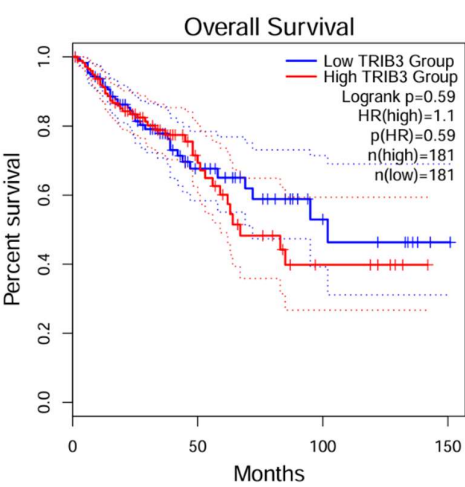
2. AXIN2



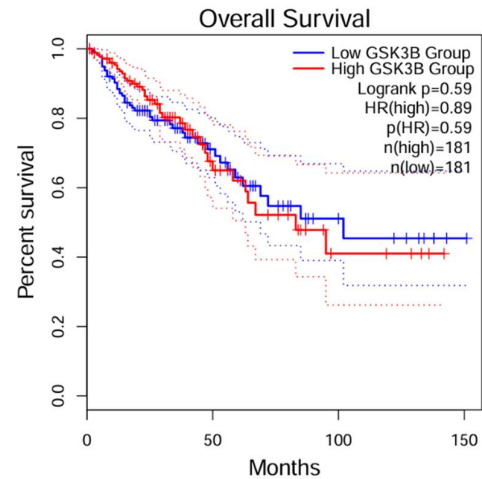
3. ENC1



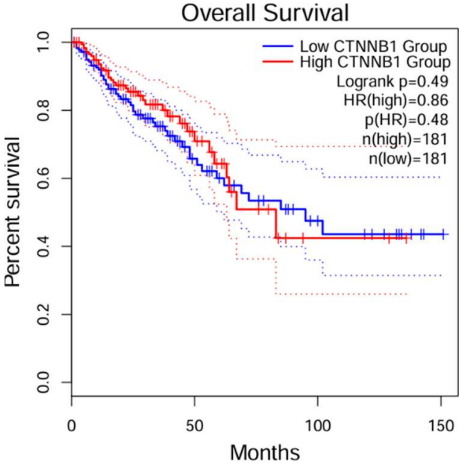
4. TRIB3



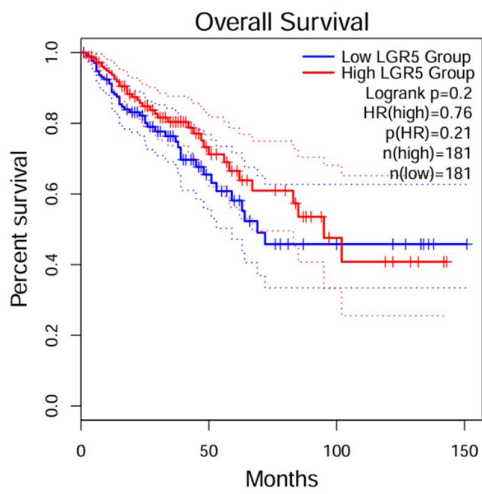
5. GSK3B



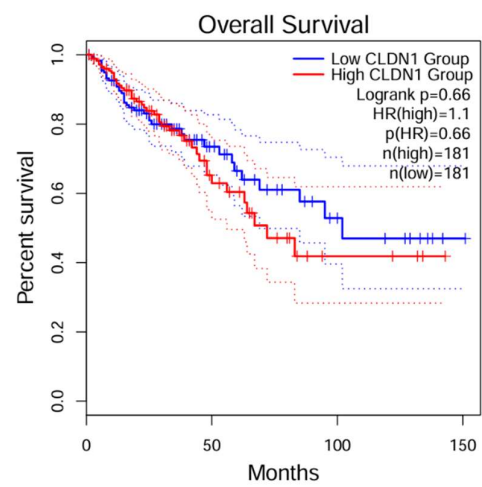
6. CTNNB1



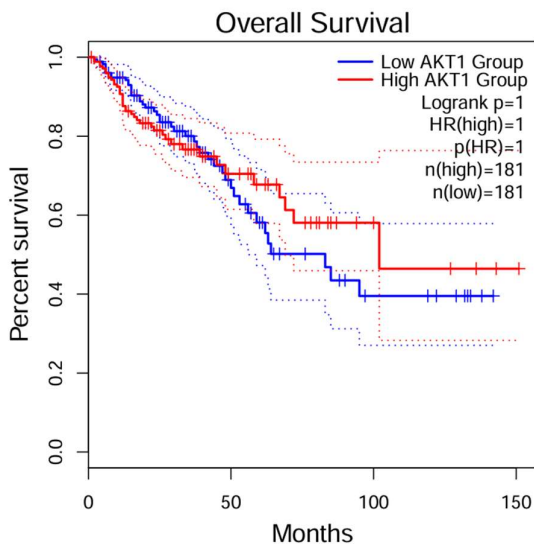
7. LGR5



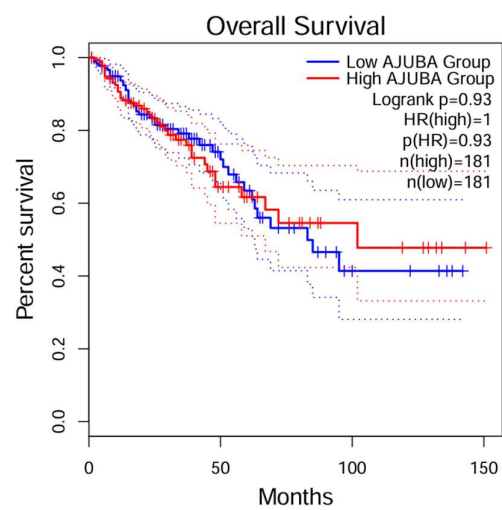
8. CLDN1



9. AKT1



10. AJUBA



(i) Identify KGs related drugs for the treatment against the disease.

Answer:

DGIdb Drug Targets 2024

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Hover each row to see the overlapping genes.

10 entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	CEP 11981	0.00001010	0.0006668	624.44	7182.69
2	EVEROLIMUS	0.0001252	0.004130	155.92	1401.11
3	SOTRASTAUIN	0.0003175	0.006909	95.86	772.11
4	IMATINIB	0.0004187	0.006909	83.04	645.92
5	CYC 116	0.001001	0.01321	52.91	365.49
6	AMG900	0.002498	0.01757	555.17	3326.77
7	GANETESPIB	0.002997	0.01757	444.11	2580.41
8	BARDOXOLONE METHYL	0.002997	0.01757	444.11	2580.41
9	MK 2206	0.002997	0.01757	444.11	2580.41
10	AZD4547	0.002997	0.01757	444.11	2580.41

Showing 1 to 10 of 66 entries | [Export entries to table](#)
Terms marked with an * have an overlap of less than 5

[Previous](#) [Next](#)

(j) Perform GO and KEGG pathway enrichment analysis of DEGs and identify top 5 GOterm from each of Biological Process (BP), Molecular Function (MF) & Cellular Component (CC) and top 5 KEGG pathways.

Answer:

GO Terms:

Biological Process (BP)

GO Biological Process 2026

Bar Graph

Table

Clustergram

Appyter



Hover each row to see the overlapping genes.

10

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entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	Positive Regulation of Proteasomal Ubiquitin-Dependent Protein Catabolic Process (GO:0032436)	3.373e-8	0.00001761	192.47	3311.50
2	Positive Regulation of Ubiquitin-Dependent Protein Catabolic Process (GO:2000060)	7.886e-8	0.00002058	154.29	2523.59
3	Positive Regulation of Proteasomal Protein Catabolic Process (GO:1901800)	1.257e-7	0.00002138	136.72	2172.45
4	Regulation of Proteasomal Ubiquitin-Dependent Protein Catabolic Process (GO:0032434)	2.047e-7	0.00002138	120.48	1855.65
5	Regulation of Protein Ubiquitination (GO:0031396)	2.047e-7	0.00002138	120.48	1855.65

Molecular Function (MF)

GO Molecular Function 2026

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Hover each row to see the overlapping genes.

10

▼

entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	Kinase Binding (GO:0019900)	0.000001554	0.00008393	42.27	565.32
2	Ubiquitin Protein Ligase Binding (GO:0031625)	0.000005854	0.0001349	50.59	609.52
3	Ubiquitin-like Protein Ligase Binding (GO:0044389)	0.000007494	0.0001349	47.44	559.91
4	Kinase Activity (GO:0016301)	0.00001216	0.0001642	92.69	1049.03
5	I-SMAD Binding (GO:0070411)	0.00002690	0.0002905	356.71	3753.85

Cellular Component (CC)

GO Cellular Component 2026

[Bar Graph](#)**Table**[Clustergram](#)[Appyter](#)

Hover each row to see the overlapping genes.

10 entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	Cell-cell Junction (GO:0005911)	0.0005166	0.009338	25.30	191.47
2	Nucleus (GO:0005634)	0.0006663	0.009338	11.02	80.61
3	Intracellular Membrane-Bounded Organelle (GO:0043231)	0.0008489	0.009338	10.54	74.50
4	Adherens Junction (GO:0005912)	0.002880	0.02376	30.60	179.00
5	Basolateral Plasma Membrane (GO:0016323)	0.004628	0.03054	23.89	128.44

KEGG pathways

KEGG 2026

[Bar Graph](#)**Table**[Clustergram](#)[Appyter](#)

Hover each row to see the overlapping genes.

10 entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	ENDOMETRIAL CANCER	4.283e-11	5.397e-9	376.17	8980.56
2	COLORECTAL CANCER	3.237e-10	2.039e-8	245.79	5370.83
3	SIGNALING PATHWAYS REGULATING PLURIPOTENCY OF STEM CELLS	3.972e-9	1.193e-7	145.99	2823.94
4	GASTRIC CANCER	4.735e-9	1.193e-7	140.77	2698.38
5	BREAST CANCER	4.735e-9	1.193e-7	140.77	2698.38

Practical 3

(a) Download the RNAseq COVID-19 dataset GSE152418 from NCBI and describes its features (Platform, BioProject No., case/control samples, No. of genes, disease name, tissue types, country names, PubMed-ID, publication date, name of organisms).

Answer:

Feature	Description
Dataset Accession No.	GSE152418
Disease Name	COVID-19 (Coronavirus Disease 2019)
Organism	Homo sapiens
Experiment Type	Expression profiling by high throughput sequencing (RNA-seq)
Platform	GPL24676 – Illumina NovaSeq 6000 (Homo sapiens)
BioProject No.	PRJNA639275
Case Samples	17 COVID-19 patient samples
Control Samples	17 healthy control samples
Total Samples	34
Tissue Type	Peripheral Blood Mononuclear Cells (PBMCs)
Country Names	Hong Kong and USA (Atlanta)
PubMed ID	32788292
Publication Title	“Systems biological assessment of immunity to mild versus severe COVID-19 infection in humans”
Publication Date	September 4, 2020
Submission Date to GEO	June 13, 2020
Platform Technology	Illumina NovaSeq 6000 RNA sequencing
Approximate No. of Genes	~60,000 transcript features (RNA-seq expression matrix)

(b) Analyze the dataset GSE152418 by using LIMMA-voom (r-package) and display the results.

(c) Identify the differentially expressed genes (DEGs) using the cutoff $P\text{-value} > 0.05$ and $|\log_2FC| > 2.0$. Separate the upregulated and downregulated DEGs.

Answer:

R Code:

```
rm(list=ls(all=TRUE))
```

```

if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")
#BiocManager::install("limma")
library(limma)
library(gclus)
library(Biobase)
library(dplyr)

getwd()
setwd("E:")
DataSet<- read.csv(file="GSE152418_raw_counts.csv", header = TRUE,
sep=",")
head(DataSet)
tail(DataSet)

ID=DataSet[,1]
head(ID)
DataS=DataSet[,-1]
rownames(DataS)<- ID
head(DataS)
dim(DataS)
Data <- voom(DataS)
#View(Data)

Data <- as.matrix(Data)
group <- factor(c(rep('Case',17),rep('control',17)))
design.mat<-model.matrix(~ group)
limma <- lmFit(Data, design.mat)
limma.e<- eBayes(limma)
results <- topTable(limma.e, coef=2, adjust="BH", n = nrow(Data),
sort.by = "none")
head(results)
write.csv(results,file="Limma_results_GSE152418.csv")

##Select Upregulated genes

Upregulated_data<- results %>% filter( adj.P.Val <0.05 & logFC >2)
dim(Upregulated_data)
head(Upregulated_data)
write.csv(Upregulated_data,"Upregulated_data_GSE152418.csv")

##Select Downregulated genes
Downregulated_data<- results %>% filter( adj.P.Val <0.05 & logFC <
-2)
dim(Downregulated_data)
head(Downregulated_data)
write.csv(Downregulated_data,"Downregulated_data_GSE152418.csv")

```

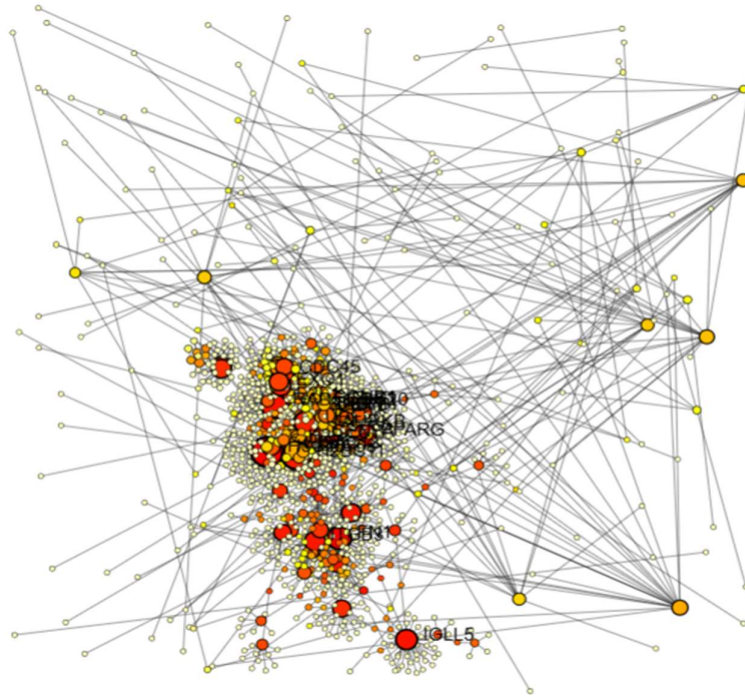
Up-Regulated Gene:

	A	B	C	D	E	F	G
1	ID	logFC	AveExpr	t	P.Value	adj.P.Val	B
2	1.05E+08	2.05001	-1.59939	4.536944	6.72E-05	0.001184	1.474809
3	134	2.161897	-1.31241	4.240722	0.00016	0.002151	0.641247
4	57282	2.523765	0.78483	5.165925	1.03E-05	0.000314	3.281624
5	728763	2.928025	1.381329	5.594449	2.85E-06	0.000128	4.527175
6	200772	3.01351	1.493091	5.093009	1.28E-05	0.000364	3.070436
7	55799	2.393051	2.060014	9.166605	9.79E-11	5.88E-08	14.47856
8	3120	2.820105	3.27052	3.798875	0.000571	0.005178	-0.56913
9	167681	2.283588	-2.85564	4.433061	9.13E-05	0.001455	1.180839
10	10842	2.448027	0.350455	3.217642	0.002827	0.01552	-2.07568
11	1.05E+08	2.487267	-1.99582	4.841481	2.72E-05	0.000613	2.34484
12	1.08E+08	2.6115	1.726566	5.872276	1.24E-06	7.16E-05	5.335974
13	10439	2.161368	1.888702	3.630241	0.000916	0.007186	-1.01773
14	3852	2.249327	-1.21858	4.37414	0.000108	0.001639	1.014858
15	28517	2.22012	0.826748	2.222564	0.002122	0.012812	-1.81256

Down-Regulated Gene:

	A	B	C	D	E	F	G
1	ID	logFC	AveExpr	t	P.Value	adj.P.Val	B
2	1.05E+08	-2.49505	-2.87006	-6.23065	4.23E-07	3.45E-05	6.37692
3	5909	-3.51011	-2.0187	-4.84683	2.68E-05	0.000606	2.360211
4	714	-2.9804	1.340396	-7.11882	3.06E-08	4.54E-06	8.924841
5	713	-2.37634	3.489206	-7.0662	3.57E-08	5.12E-06	8.775686
6	6007	-2.64239	0.793249	-3.74366	0.000667	0.005772	-0.71694
7	63967	-2.31278	1.673647	-9.79058	1.89E-11	1.65E-08	16.05905
8	991	-3.33921	2.52251	-11.2301	5.12E-13	2.04E-09	19.49291
9	6536	-2.70243	-0.85305	-6.16383	5.17E-07	3.89E-05	6.183179
10	11004	-2.57663	1.91933	-9.74685	2.11E-11	1.81E-08	15.9501
11	4998	-2.19358	1.423212	-7.08222	3.41E-08	4.94E-06	8.821112
12	200132	-2.36698	-2.12554	-5.11787	1.19E-05	0.000345	3.142404
13	55635	-3.2855	-0.74354	-9.53267	3.70E-11	2.75E-08	15.41252
14	51375	-2.12917	-0.90944	-3.88167	0.000451	0.004412	-0.34595
15	2949	-2.36008	-1.33417	-6.06951	6.85E-07	4.74E-05	5.909417
16	8991	-4.34022	-0.23445	-4.24776	0.000157	0.002121	0.660863
17	3782	-3.73994	0.567028	-9.12331	1.10E-10	6.19E-08	14.36685

Answer: PPI for All Network:

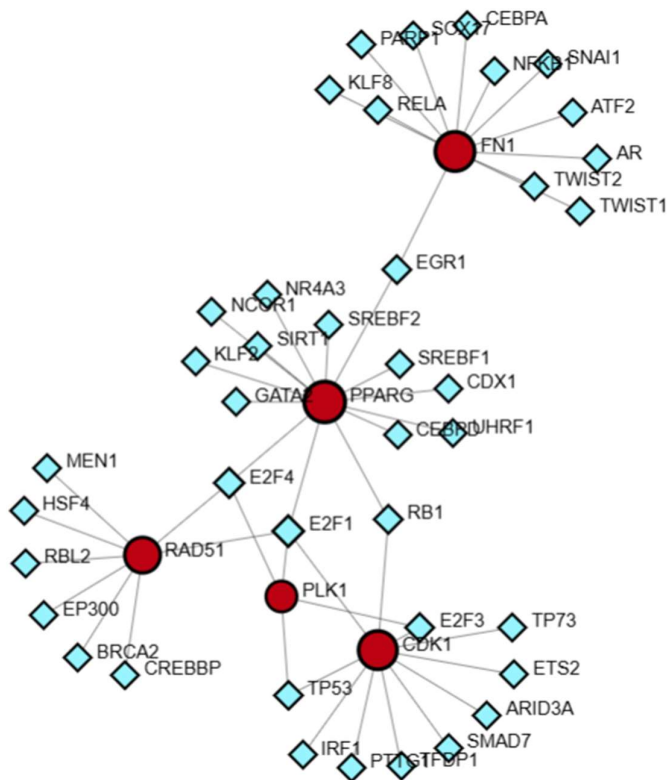


Top 10 Key Genes:

Id	Label	Degree	Betweenness	Expression	Rank D	Rank B	Rank A
8357	H3C10	121	97279.34	0	1	2	1.5
983	CDK1	99	93438.08	0	2	3	2.5
1058	CENPA	86	75724	0	5	4	4.5
5347	PLK1	78	62234.78	0	6	6	6
2335	FN1	56	113984.68	0	14	1	7.5
5888	RAD51	66	55335.93	0	9	7	8
3012	H2AC8	97	35688.36	0	3	20	11.5
8970	H2BC11	95	37175.1	0	4	19	11.5
100423062	IGLL5	42	54253.65	0	17.5	8	12.75
5468	PPARG	48	46785.5	0	15	11	13

(e) Construct the TFs-KGs interaction network and identify 5 key TFs as the transcription regulators of KGs.

Answer: **TFs-KGs Network:**



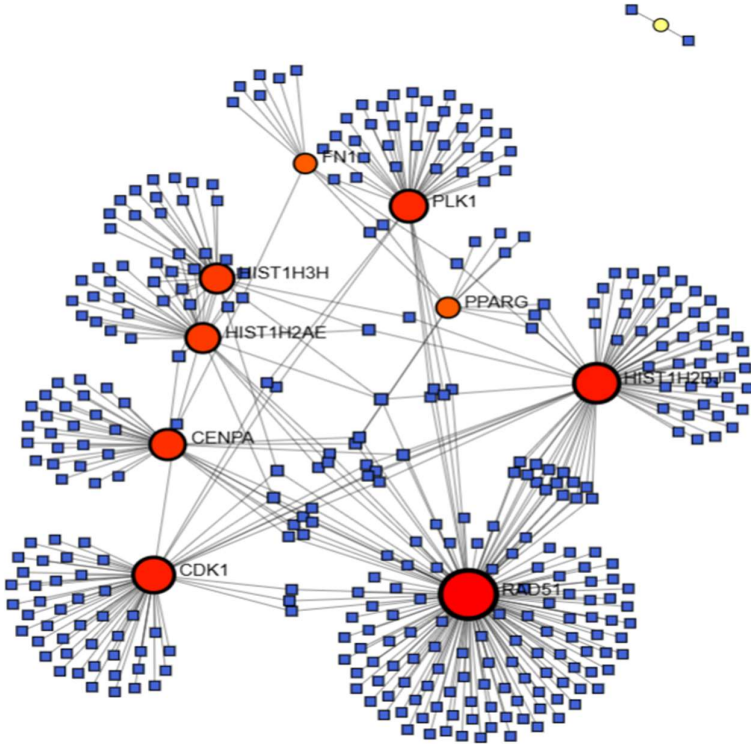
Top 5 TFs:(Yellow cells)

	A	B	C	D	E	F	G	H
1	Id	Label	Degree	Betweenness	Expression	Rank D	Rank B	Rank A
2	5468	PPARG	14	667.57	0	1	1	1
3	2335	FN1	12	418	0	2	2	2
4	983	CDK1	11	315.6	0	3	4	3.5
5	5888	RAD51	8	244.67	0	4	6	5
6	1869	E2F1	4	296.57	0	5.5	5	5.25
7	1958	EGR1	2	384	0	9.5	3	6.25
8	1874	E2F4	3	118.7	0	7	7	7
9	5347	PLK1	4	41.17	0	5.5	9	7.25
10	5925	RB1	2	109.33	0	9.5	8	8.75
11	1871	E2F3	2	4.2	0	9.5	10.5	10
12	7157	TP53	2	4.2	0	9.5	10.5	10

(f) Construct the miRNA-KGs interaction network and identify 5 key miRNA as the posttranscription regulators of KGs.

Answer:

miRNA-KG Network:



Top 5 Key miRNA: (Yellow Cells)

	A	B	C	D	E	F	G	H
1	Id	Label	Degree	Betweenness	Expression	Rank D	Rank B	Rank A
2	5888	RAD51	147	45311.95	0	1	1	1
3	8970	HIST1H2BJ	73	20747.35	0	2	2	2
4	983	CDK1	55	16449.29	0	3	3	3
5	5347	PLK1	40	12090.71	0	4	4	4
6	1058	CENPA	35	9586.93	0	5	5	5
7	3012	HIST1H2AE	31	6860.7	0	6.5	6	6.25
8	8357	HIST1H3H	31	6826.38	0	6.5	7	6.75
9	MIMAT0000255	hsa-miR-34a-5p	4	3910.68	0	10	8	9
10	5468	PPARG	11	2387.45	0	8	12	10
11	MIMAT0002819	hsa-miR-193b-3p	3	3593.83	0	14	9	11.5
12	MIMAT0000092	hsa-miR-92a-3p	3	3185.11	0	14	10	12
13	MIMAT0015073	hsa-miR-3190-5p	3	2803.76	0	14	11	12.5
14	2335	FN1	10	2317.25	0	9	17	13
15	MIMAT0000222	hsa-miR-192-5p	3	1610.84	0	14	19.5	16.75
16	MIMAT0000272	hsa-miR-215-5p	3	1610.84	0	14	19.5	16.75
17	MIMAT0005930	hsa-miR-1276	3	1373.13	0	14	24	19
18	MIMAT0022486	hsa-miR-5693	3	1162.46	0	14	27	20.5

(g) Construct the diseases-KGs interaction network and identify top 10 diseases risks as comorbidities associated with KGs.

Answer:

Top 10 Diseases linked to KGs:

DisGeNET

Bar Graph

Table

Clustergram

Appyter

⚙️ ⓘ

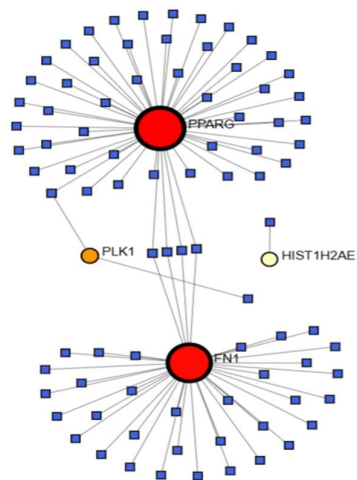
Hover each row to see the overlapping genes.

10 entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	Triple Negative Breast Neoplasms	1.833e-7	0.0001708	46.40	719.76
2	Hepatoblastoma	0.000003691	0.001378	57.02	713.35
3	Carcinoma, Large Cell	0.000004434	0.001378	131.37	1619.33
4	Lymphoma	0.00001287	0.002998	21.55	242.64
5	Ataxia Telangiectasia	0.00002892	0.005391	33.33	348.33
6	Lip and Oral Cavity Carcinoma	0.00005793	0.007867	27.75	270.72
7	Adenocarcinoma	0.00006053	0.007867	16.08	156.14
8	Malignant neoplasm of mouth	0.00006753	0.007867	26.64	255.84
9	Osteosarcoma of bone	0.00007690	0.007963	18.28	173.14
10	Hematopoietic Neoplasms	0.00009946	0.008211	44.90	413.79

Disease KG Network:



(h) Perform survival analysis with KGs and interpret the results.

Currently we don't have the facility to do survival analysis on COVID 19.

Only cancer data is available for survival analysis at present.

(i) Identify KGs related drugs for the treatment against the disease.

DGIdb Drug Targets 2024

Bar GraphTableClustergramAppyter⚙️📘

Hover each row to see the overlapping genes.

10▼ entries per page

Search:

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	SOTRASTAUIN	0.0003175	0.01334	95.86	772.11
2	RG 1530	0.0008042	0.01689	59.24	422.16
3	BARDOXOLONE METHYL	0.002997	0.01860	444.11	2580.41
4	DINACICLIB	0.002997	0.01860	444.11	2580.41
5	RGB 286638	0.003495	0.01860	370.07	2093.27
6	AMUVATINIB	0.003495	0.01860	370.07	2093.27
7	BAY1000394	0.004492	0.01860	277.53	1500.17
8	CINNARIZINE	0.004990	0.01860	246.68	1307.49
9	DACARBAZINE	0.004990	0.01860	246.68	1307.49
10	ALVOCIDIB	0.004990	0.01860	246.68	1307.49

Showing 1 to 10 of 42 entries | [Export entries to table](#)

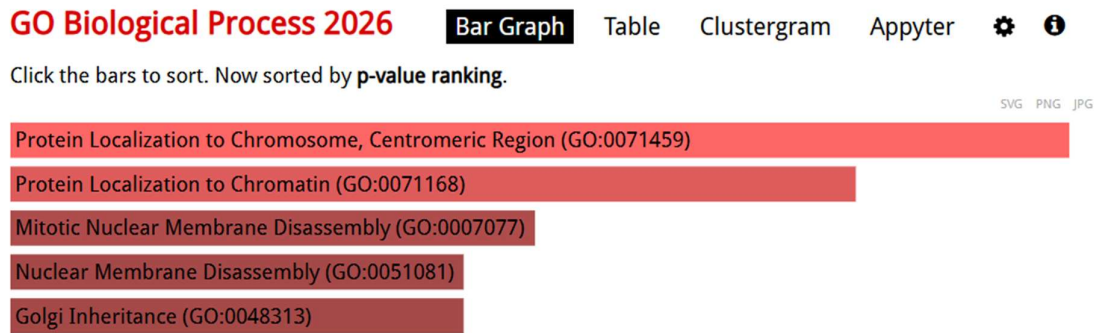
Terms marked with an * have an overlap of less than 5

PreviousNext

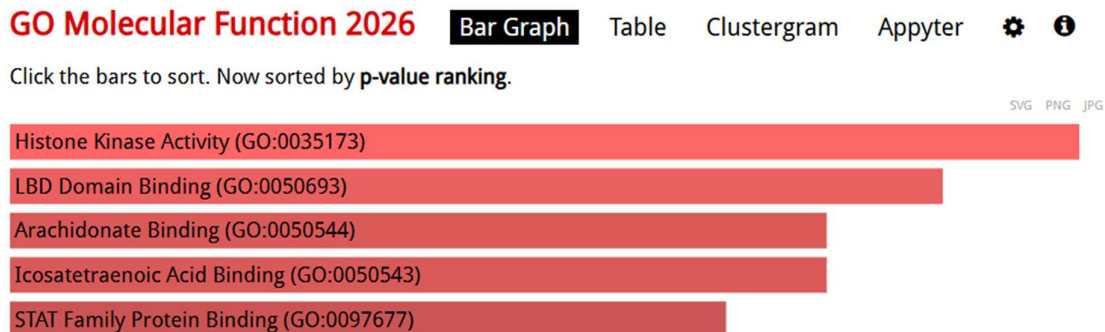
(j) Perform GO and KEGG pathway enrichment analysis of DEGs and identify top 5 GO-term from each of Biological Process (BP), Molecular Function (MF) & Cellular Component (CC) and top 5 KEGG pathways.

Answer:

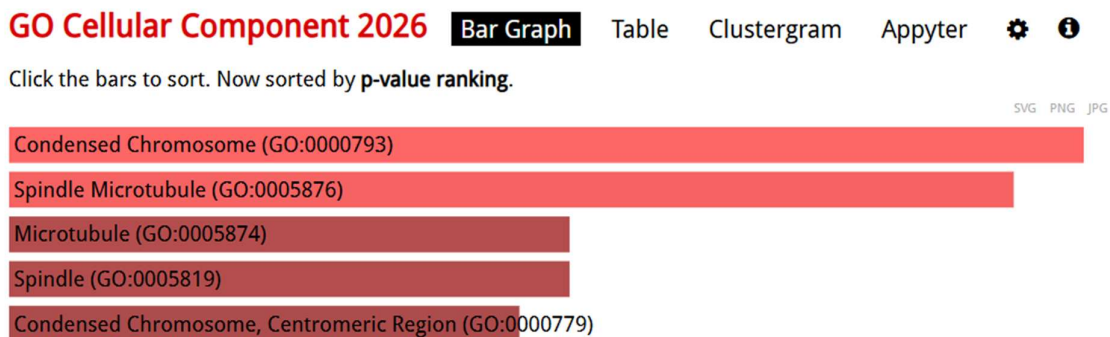
Biological Process (BP):



Molecular Function (MF):



Cellular Component (CC):



5 KEGG pathways:

